

- *30. A uniformly charged sphere of radius a is surrounded by a uniformly charged concentric spherical shell of inner radius b and outer radius c (Fig. 25.35). The total charge in the sphere is Q , and that on the outer shell $-Q$. Find the potential at $r = b$, at $r = a$, and at $r = 0$.

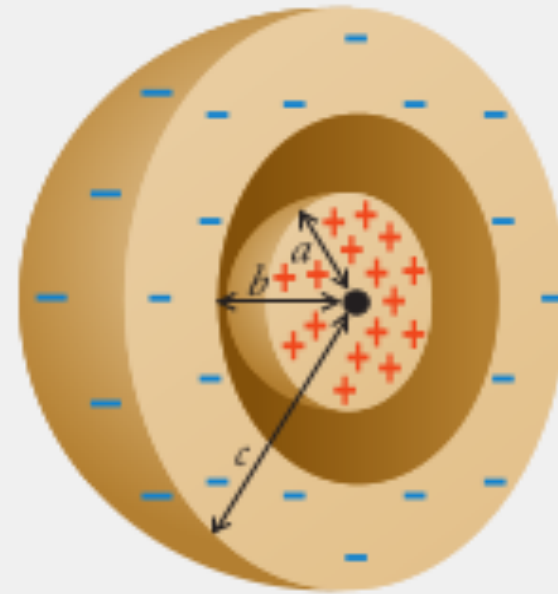


FIGURE 25.35 Charged sphere and concentric spherical shell.

46. In a region of space, the electrostatic potential is described by

$$V = x^2y + 3xyz + zy^2$$

Find the electric field in this region.

73. A disk of radius R (Fig. P25.73) has a nonuniform surface charge density $\sigma = Cr$, where C is a constant and r is measured from the center of the disk to a point on the surface of the disk. Find (by direct integration) the electric potential at P .

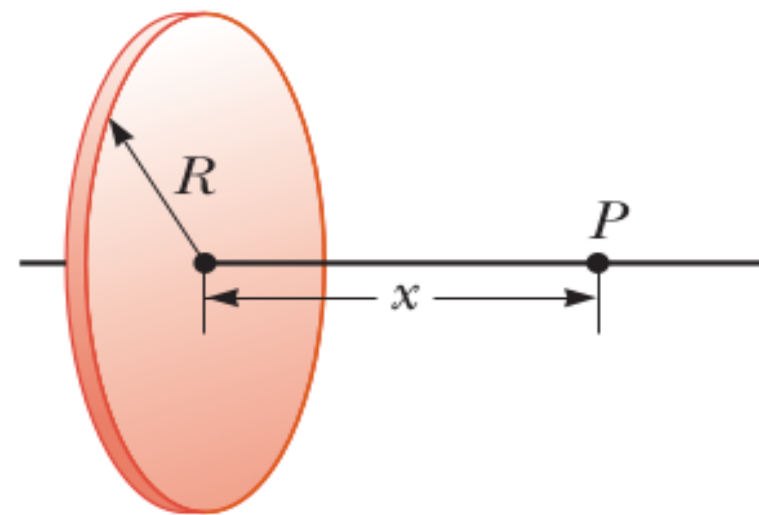


Figure P25.73

*50. Two rods of equal lengths l form a symmetric cross. The rods carry charges $\pm Q$ uniformly distributed along their lengths. Calculate the potential at the point P at a distance x from one end of the cross (Fig. 25.38). Calculate the electric field at this point.

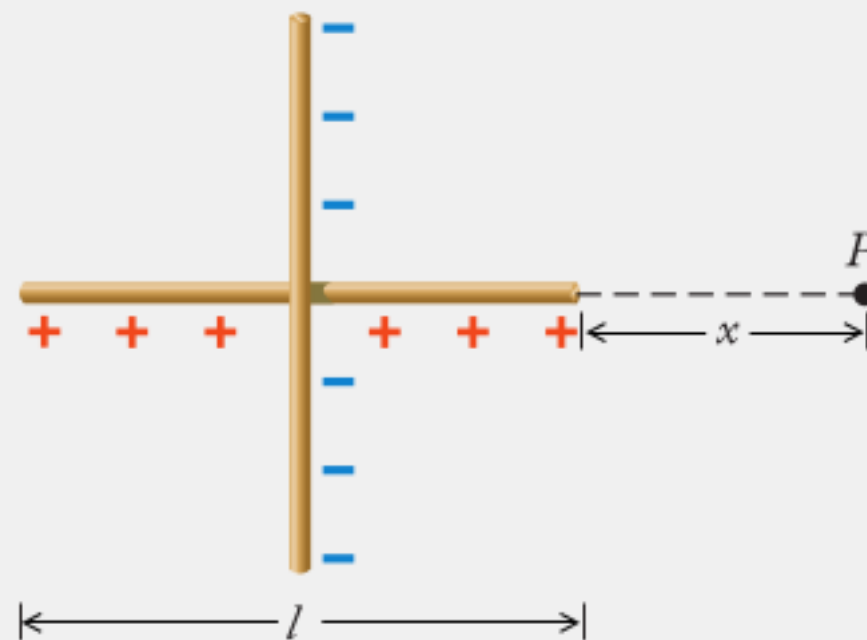


FIGURE 25.38 Two charged rods forming a cross.

- *28. A total charge Q is distributed uniformly along a straight rod of length l . Find the potential at a point P a distance h from the midpoint of the rod (see Fig. 25.34).

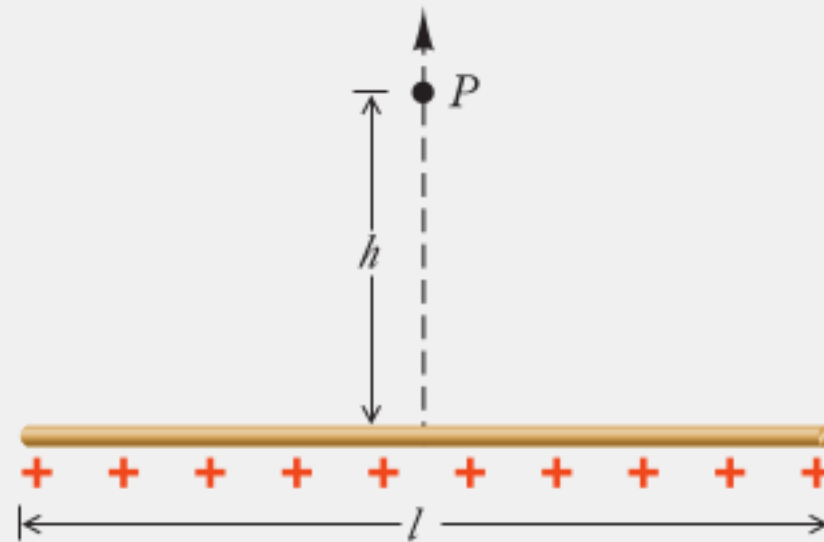


FIGURE 25.34 A uniformly charged rod of length l .

- *24. Three large charged sheets are parallel to the x - z plane. The sheets are at $y = 0$, $y = d$, and $y = 2d$; these sheets have uniform surface charge densities $+\sigma$, -2σ , and $+\sigma$, respectively. Assuming the reference potential is zero at $y = 0$, determine the potential as a function of y .